Name: ____________ Cohort/Batch: ____________ Date: ____________ Score: ____________

# BLOB STORAGE PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Cloud

TOTAL: 10 QUESTIONS

**Q1** What is Blob Storage and what is its primary purpose in its cloud ecosystem? *Threat Model*

**Q2** How does IAM work with Blob Storage? *Architecture*

**Q3** How do you monitor Blob Storage in production? *Configuration*

**Q4** How do you scale Blob Storage to handle variable load? *Hardening*

**Q5** How do you secure Blob Storage in production? *SSO / IdP*

**Q6** How do you manage configuration and secrets for Blob Storage? *Observability*

**Q7** What are the main cost drivers for Blob Storage and how do you optimize them? *Lifecycle*

**Q8** How do you implement high availability with Blob Storage? *Compliance*

**Q9** How does Blob Storage integrate with a CI/CD pipeline? *Testing*

**Q10** What are common pitfalls when first adopting Blob Storage? *Tuning*

# ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

### A1 Blob Storage is a managed cloud

Mitigation

Blob Storage is a managed cloud service that handles a specific infrastructure concern (compute, storage, messaging, AI). Using it shifts operational burden to the cloud provider while you focus on application logic.

### A2 Access to Blob Storage is controlled

Primitives

Access to Blob Storage is controlled via IAM roles and policies. Attach the minimum required permissions to a service account or role. Avoid long-lived access keys; use instance roles or Workload Identity Federation instead.

### A3 Blob Storage emits metrics to the

Hardening

Blob Storage emits metrics to the cloud provider's monitoring service (CloudWatch, Cloud Monitoring, Azure Monitor). Set alerts on key metrics (error rate, latency, capacity). Use distributed tracing to correlate across services.

### A4 Blob Storage supports auto-scaling policies based

Gotchas

Blob Storage supports auto-scaling policies based on CPU, queue depth, or custom metrics. Set minimum and maximum capacity. Horizontal scaling (more instances) is generally preferred over vertical for cloud workloads.

### A5 Place Blob Storage in a private

Federation

Place Blob Storage in a private subnet with no public endpoint where possible. Use VPC security groups or firewall rules to allow only required traffic. Encrypt data at rest and in transit. Rotate credentials regularly.

### A6 Store sensitive values in a managed

Observability

Store sensitive values in a managed secret store (AWS Secrets Manager, GCP Secret Manager, Azure Key Vault). Inject secrets as environment variables at runtime, never bake them into container images or source code.

### A7 Cost in Blob Storage typically comes

Automation

Cost in Blob Storage typically comes from compute hours, data transfer, storage, and request counts. Right-size instances, use reserved capacity for steady-state workloads, and enable lifecycle policies to expire old data.

### A8 Deploy Blob Storage across multiple availability

Governance

Deploy Blob Storage across multiple availability zones. Use managed replicas or active-active configurations. Implement health checks and automatic failover. Test resilience regularly with chaos engineering or failover drills.

### A9 CI/CD pipelines use the cloud CLI

Assessment

CI/CD pipelines use the cloud CLI or SDK to deploy updates to Blob Storage after tests pass. Infrastructure-as-code (Terraform, CDK) defines Blob Storage configuration as version-controlled code deployed via the pipeline.

### A10 Common mistakes include misconfigured IAM

Optimization

Common mistakes include misconfigured IAM policies (too permissive), no cost alerting, deploying only in one availability zone, and missing backups. Read the service SLA carefully to understand what the cloud provider guarantees.